PROJECT REPORT

AI Financial Advisor

1. INTRODUCTION

1.1 Project Overview

The AI Financial Advisor is an intelligent web-based system designed to provide personalized financial guidance using artificial intelligence. The platform analyzes user financial data such as income, expenses, savings, and investment goals to generate tailored financial advice. It helps individuals make better decisions about budgeting, saving, investing, and financial planning.

The system uses AI models and data analytics to simulate the role of a professional financial advisor. By automating financial planning, the platform makes financial advice accessible to a larger audience, especially individuals who cannot afford professional financial consultants.

The application is deployed online and allows users to interact through a simple interface where they input financial information and receive instant recommendations.

1.2 Purpose

The purpose of this project is to develop an intelligent platform that provides automated financial advice and planning support.

The system aims to:

- Provide personalized financial recommendations
- Help users manage income, expenses, and savings
- Improve financial literacy
- Support investment and budgeting decisions
- Make financial advisory services accessible to everyone

2. IDEATION PHASE

2.1 Problem Statement

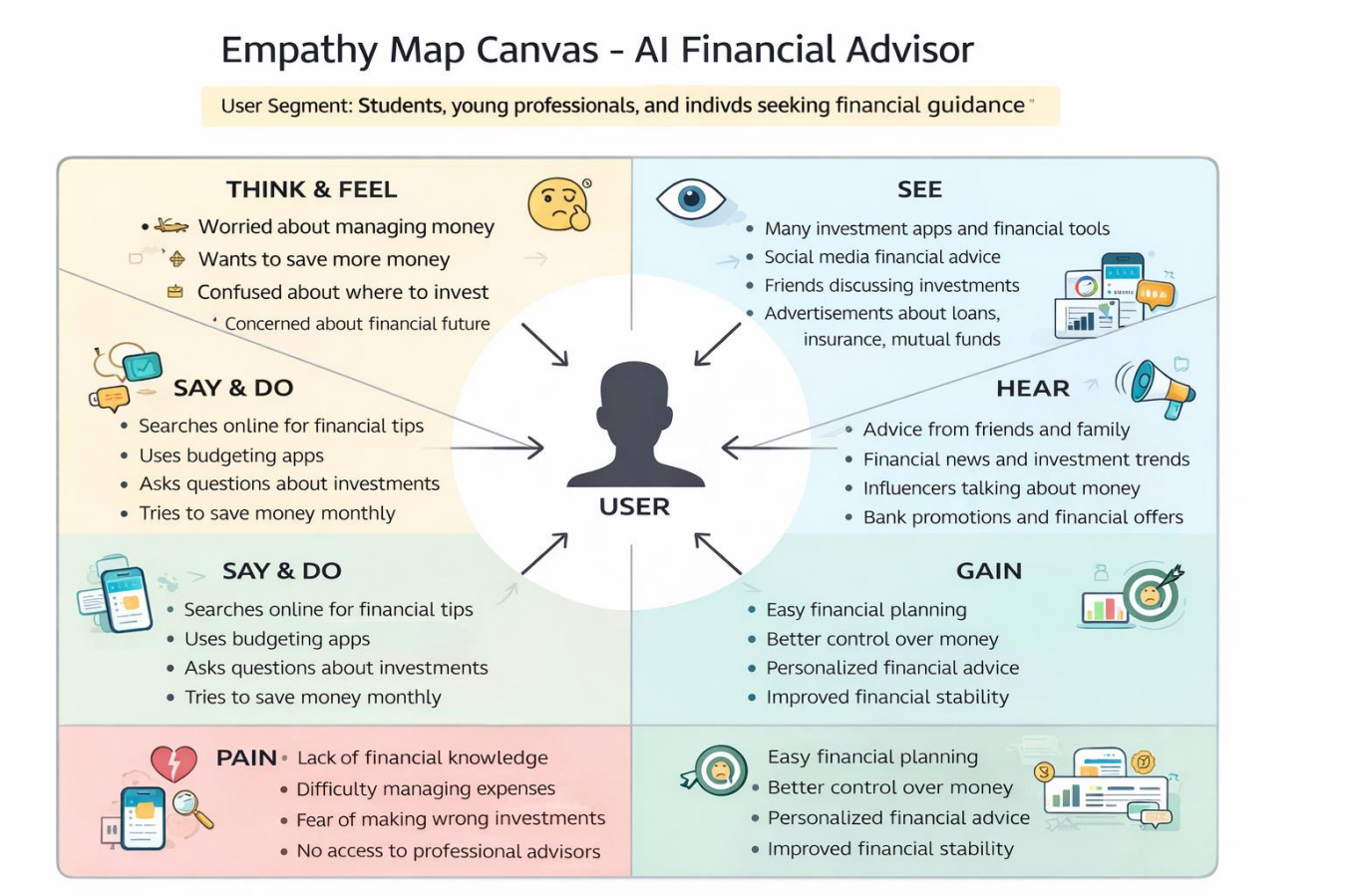
Many individuals struggle with financial planning due to a lack of knowledge, time, or access to professional advisors. Traditional financial advisory services are often expensive and inaccessible to common users.

Common financial challenges include:

- Poor budgeting habits
- Lack of investment knowledge
- Difficulty managing savings and expenses
- Limited access to financial experts
- Lack of personalized financial advice

Therefore, there is a need for an AI-powered platform that can provide affordable, personalized, and automated financial guidance.

2.2 Empathy Map Canvas

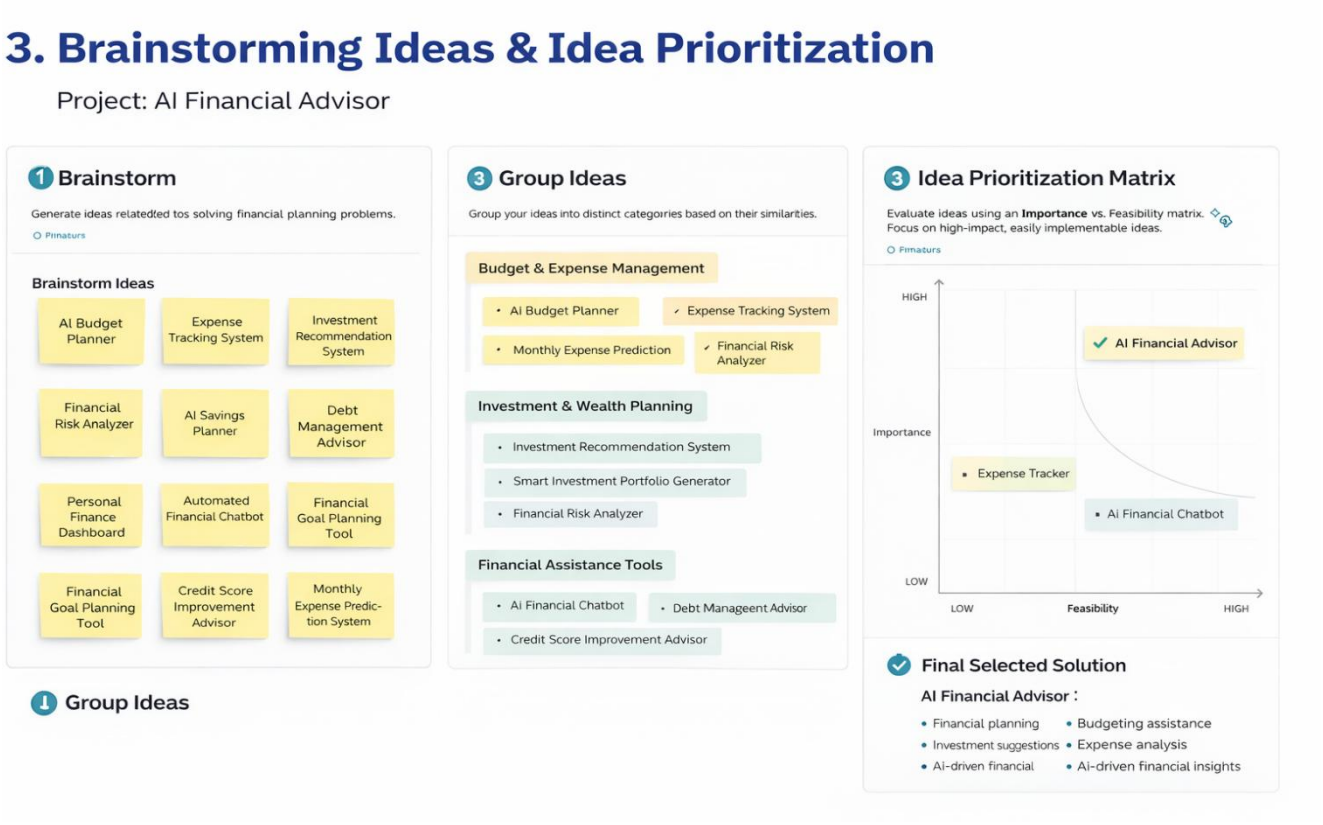


2.3 Brainstorming

During brainstorming, several possible solutions were considered:

- Budget tracking system
- Investment recommendation platform
- Expense analyzer using AI
- Personal financial planning assistant

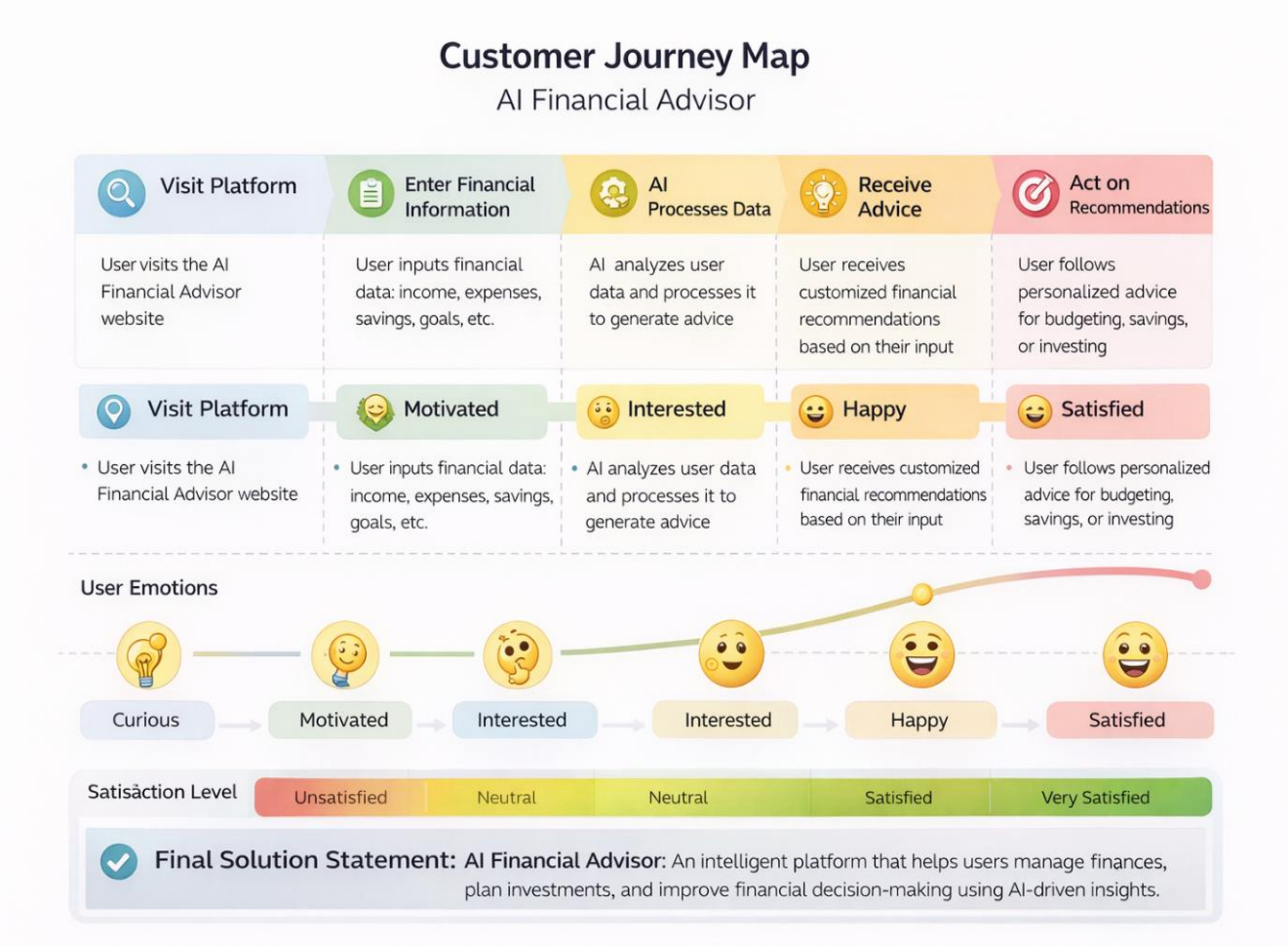
After evaluating these ideas, the final solution selected was an AI Financial Advisor, which combines budgeting, financial planning, and AI-driven insights.



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User visits the platform
2. User enters financial information
3. System processes data using AI algorithms
4. Personalized financial advice is generated
5. User receives recommendations for budgeting, savings, and investment



3.2 Solution Requirement

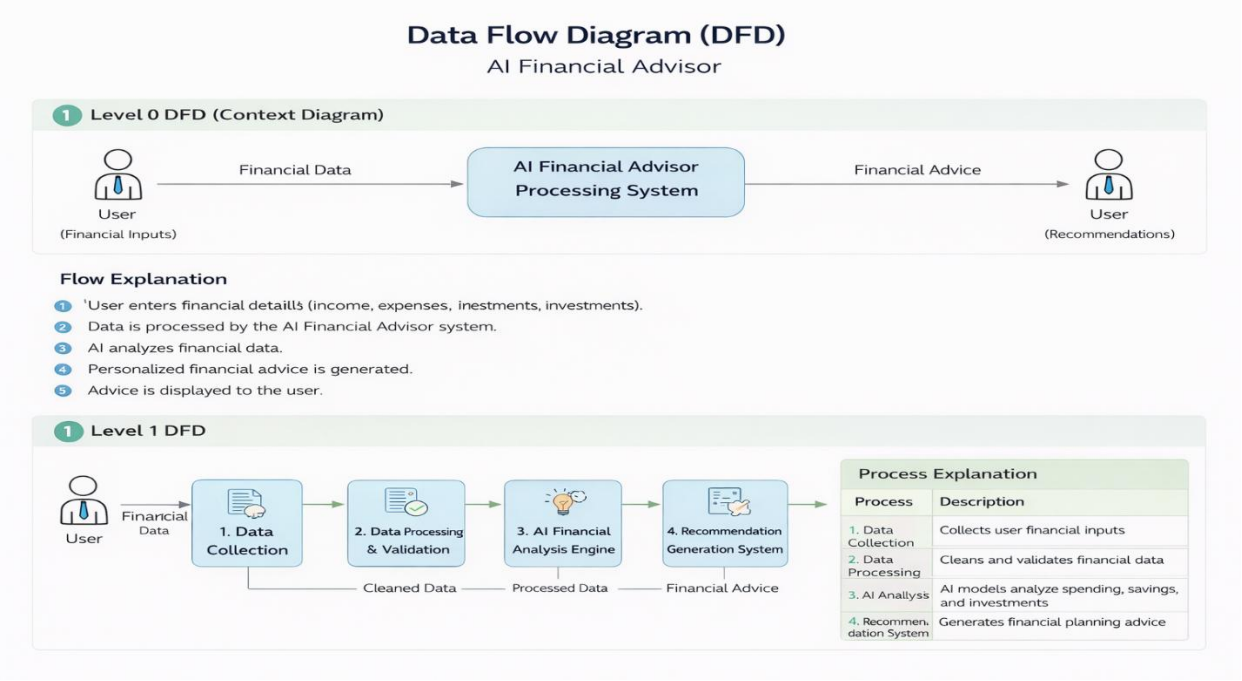
Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form
		Registration using Email
		Registration through Google Account
FR-2	User Login & Authentication	Login using Email and Password
		Secure authentication system
		Password reset option
FR-3	Financial Data Input	User enters income details
		User enters expense details
		User enters savings and financial goals
FR-4	Financial Data Analysis	AI analyzes financial information
		Process user financial patterns
		Identify spending habits
FR-5	Personalized Financial Advice	Generate AI-based financial recommendations
		Provide budgeting suggestions
		Provide saving strategies
FR-6	Investment Recommendation	Suggest investment options
		Risk-based investment guidance
		Long-term financial planning suggestions
FR-7	Dashboard & Results Display	Display financial insights
		Show recommendations visually
		Provide financial summary

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a simple and user-friendly interface so users can easily input financial information and understand the advice.
NFR-2	Security	The system should ensure secure handling of user financial data using authentication and data protection methods.
NFR-3	Reliability	The platform should provide accurate financial recommendations and operate consistently without system failures.
NFR-4	Performance	The system should process user financial data and generate recommendations quickly with minimal delay.
NFR-5	Availability	The AI Financial Advisor platform should be accessible online at any time for users.
NFR-6	Scalability	The system should support increasing numbers of users without performance issues.

3.3 Data Flow Diagram



User Stories Table

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Mobile User)	Registration	USN-1	As a user, I can register using email and password	User can successfully create account	High	Sprint-1
Customer (Mobile User)	Login	USN-2	As a user, I can login using email and password	User can access dashboard	High	Sprint-1
Customer	Financial Data Entry	USN-3	As a user, I can enter my income and expenses	System saves financial data	High	Sprint-2
Customer	AI Analysis	USN-4	As a user, I can receive financial insights	AI generates analysis	Medium	Sprint-2
Customer	Recommendations	USN-5	As a user, I can view financial advice	Recommendations are shown on dashboard	High	Sprint-3
Customer	Dashboard	USN-6	As a user, I can view financial reports	Graphs and summaries displayed	Medium	Sprint-3
Admin	System Monitoring	USN-7	Admin can monitor system performance	Admin dashboard available	Low	Sprint-4

3.3 Technology Stack

Table: Technology Components

S.No	Component	Description	Technology
1	User Interface	Interface through which users interact with the financial advisor system via web browser	HTML, CSS, JavaScript
2	Frontend Framework	Handles UI rendering, forms, and user interaction	HTML5, CSS3, JavaScript
3	Application Logic	Core logic of the system for processing user financial inputs and generating advice	Python
4	Backend Framework	Handles API requests, routing, and communication between frontend and AI modules	Flask / FastAPI
5	AI Processing Module	Performs financial analysis and generates personalized financial advice	Machine Learning Algorithms
6	Data Processing	Processes user financial data such as income, expenses, savings, and investments	Python Data Processing Libraries
7	Database	Stores user financial information and system data securely	SQLite / NoSQL Database
8	Cloud Deployment	Hosts the web application and makes it accessible online	Render Cloud Platform
9	API Integration	Handles communication between frontend and backend services	REST APIs
10	Infrastructure	Server environment used to deploy and run the application	Cloud-based Web Server

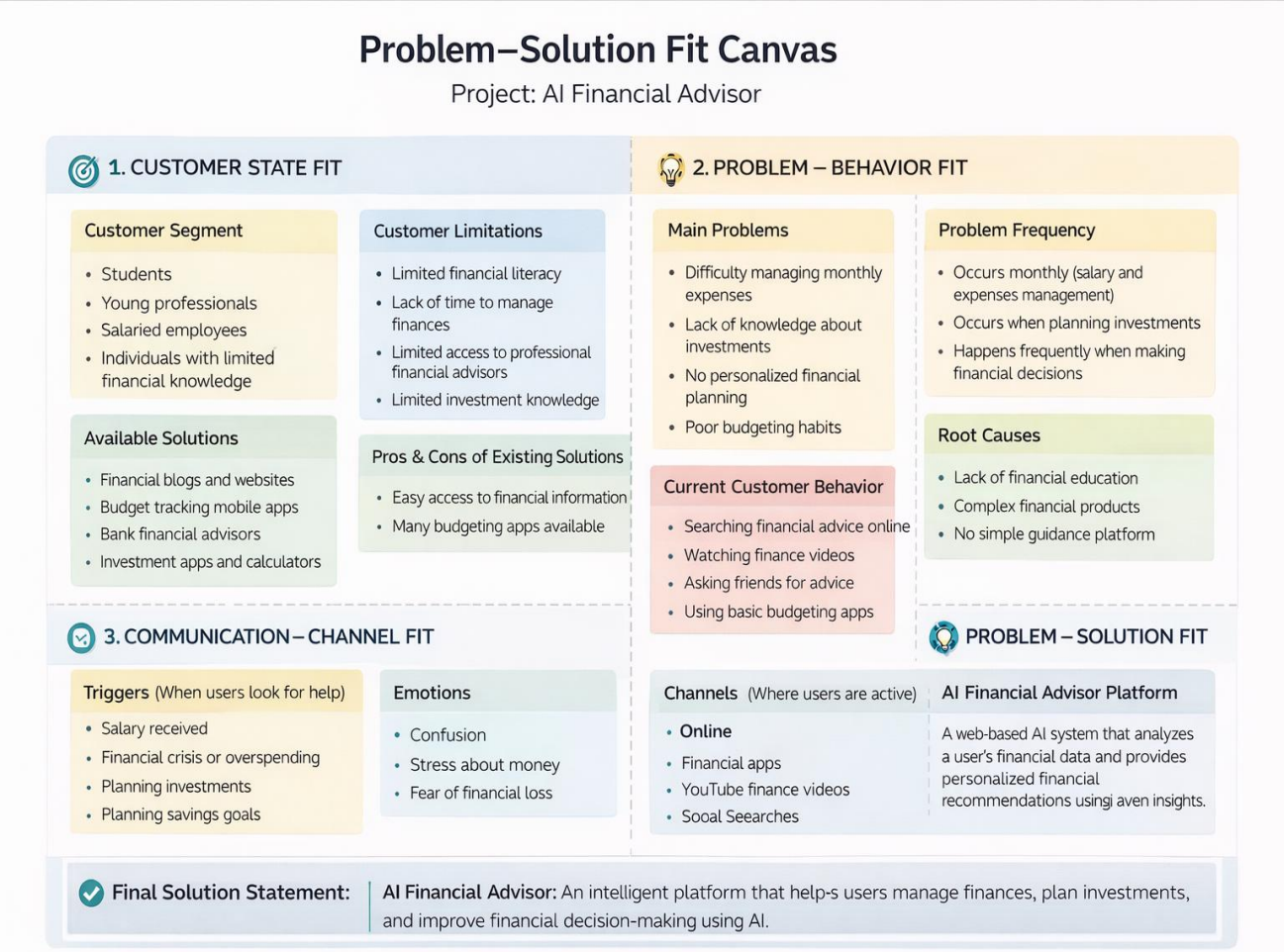
Table: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Use of open-source technologies for development	Python, Flask/FastAPI
2	Security Implementations	Secure handling of financial data and user information	HTTPS, Secure API Calls
3	Scalable Architecture	System designed to support increasing users and financial data	Cloud Deployment on Render
4	Availability	Application accessible online for users anytime	Web-based Cloud Hosting

4. PROJECT DESIGN

4.1 Problem Solution Fit

The AI Financial Advisor addresses the problem of limited access to financial experts by providing an automated solution. The system offers personalized advice based on individual financial data, making financial planning simple and accessible.

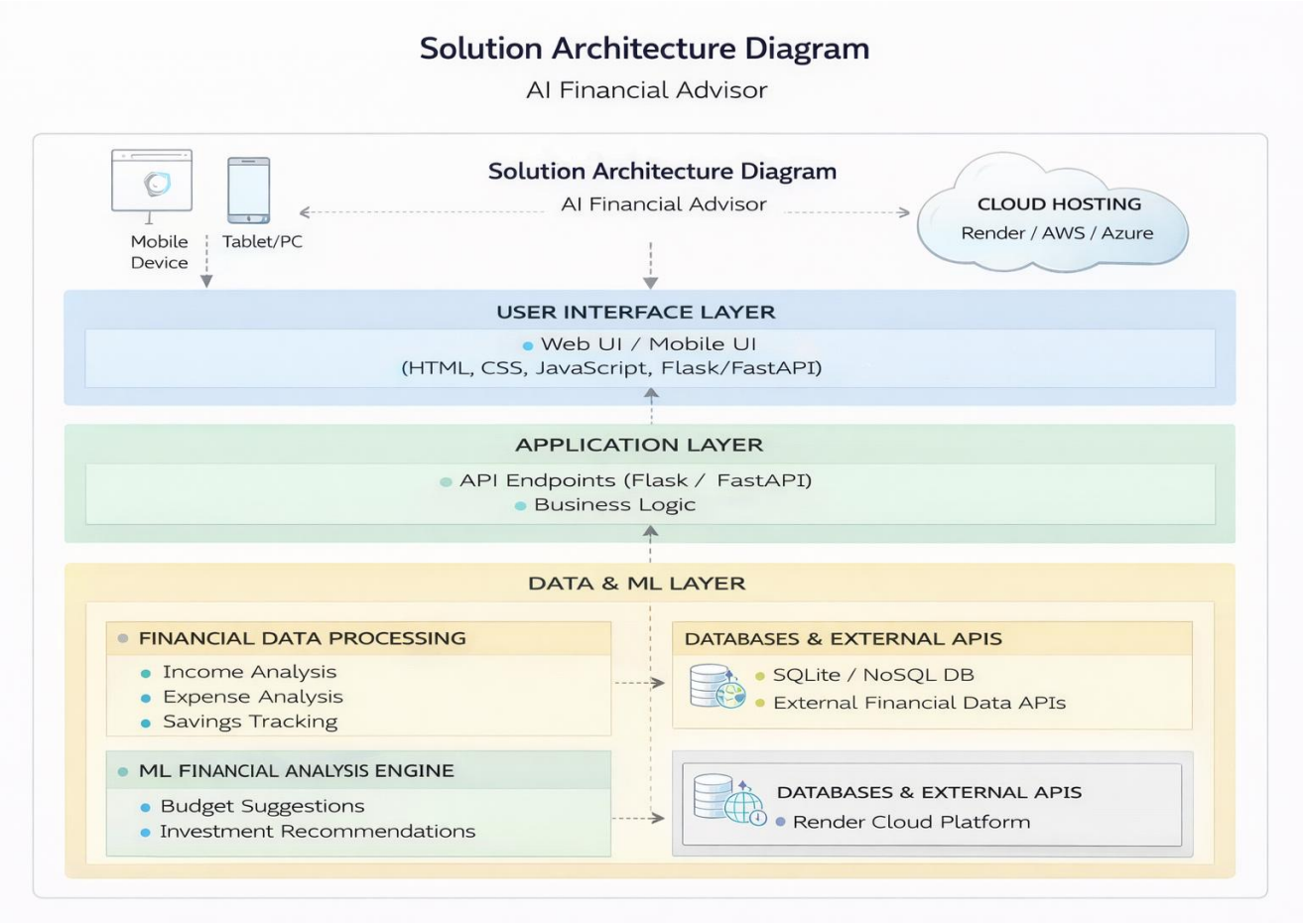


4.2 Proposed Solution

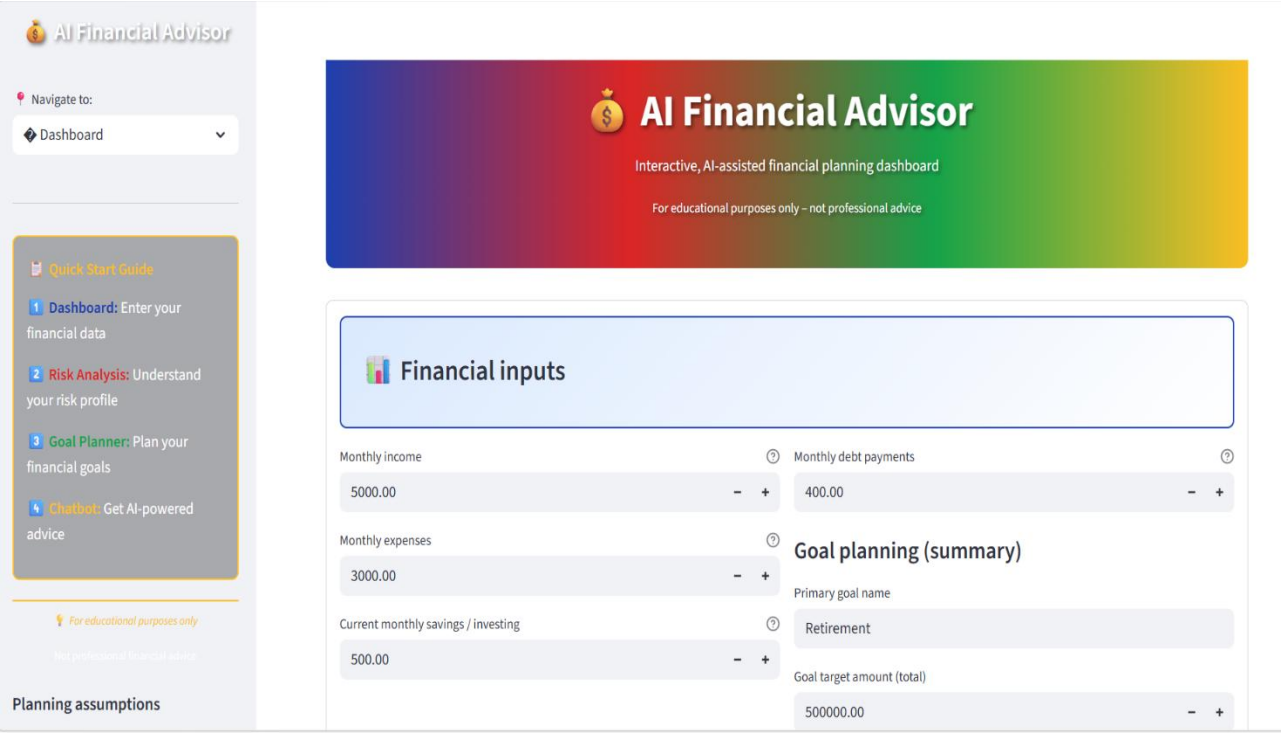
The proposed solution is a web-based AI platform that allows users to input their financial details and receive personalized financial recommendations.

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	Many individuals lack proper financial knowledge and guidance to manage their income, expenses, savings, and investments effectively. Traditional financial advisors are expensive and not easily accessible to everyone. As a result, people often make poor financial decisions, leading to financial instability.
2	Idea / Solution Description	The proposed solution is an AI Financial Advisor, a web-based platform that uses artificial intelligence to analyze user financial data and provide personalized financial advice. Users can input details such as income, expenses, savings, and financial goals. The system processes this information and generates recommendations for budgeting, savings, and investment planning.
3	Novelty / Uniqueness	The uniqueness of this solution lies in using AI-driven financial analysis to provide automated, personalized financial guidance. Unlike traditional advisory services, the platform is accessible online, cost-effective, and available 24/7, making financial planning easier for individuals with limited financial knowledge.
4	Social Impact / Customer Satisfaction	This solution helps individuals improve their financial literacy and decision-making. By providing personalized financial insights, users can manage their finances better, reduce unnecessary expenses, increase savings, and plan investments effectively. This leads to improved financial stability and better financial awareness in society.
5	Business Model (Revenue Model)	The platform can generate revenue through subscription plans, premium financial insights, partnerships with financial institutions, and targeted financial product recommendations such as investment plans or insurance services.
6	Scalability of the Solution	The solution is designed as a cloud-based web platform, making it scalable for a large number of users. With cloud infrastructure and AI models, the system can easily support increasing users and financial data without major changes to the architecture.

4.3 Solution Architecture



7. RESULTS


AI Financial Advisor
Navigate to:
Dashboard
Quick Start Guide
1 Dashboard: Enter your financial data
2 Risk Analysis: Understand your risk profile
3 Goal Planner: Plan your financial goals
4 Chatbot: Get AI-powered advice
For educational purposes only
Planning assumptions
AI Financial Advisor
Interactive, AI-assisted financial planning dashboard
For educational purposes only – not professional advice
Financial inputs
Monthly income
5000.00
Monthly debt payments
400.00
Monthly expenses
3000.00
Current monthly savings / investing
500.00
Goal planning (summary)
Primary goal name
Retirement
Goal target amount (total)
500000.00

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For educational purposes only
Snapshot of your financial health
Savings rate
10.0%
Expense ratio
60.0%
Debt-to-income (DTI)
8.0%
Free cash flow
$1,600.00
Financial health score
48/100
Health Grade: D
Budget breakdown
Portion of income
60%
50%
40%
30%
20%
10%
0%
Needs
Savings
Wants
Category
Category
Needs
Savings
Wants
Income vs. Debt payments
Monthly amount
5,000
4,000
3,000
2,000
1,000
0
Debt payments
Income
Type
Debt payments
Income

AI Financial Advisor

Navigate to:

Dashboard

Quick Start Guide

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Emergency fund & risk profile

Emergency Fund Target

For **6 months** of expenses, your recommended emergency fund is approximately:

$18,000.00

Risk Profile

Risk score: 50/100 (Medium)

Balanced tolerance for risk and return.

AI Financial Advisor

Navigate to:

Risk analysis

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Financial Health & Risk Analysis

Understand your financial strengths and areas for improvement

Key Financial Metrics

Savings rate	Expense ratio	Debt-to-income (DTI)	Free cash flow
10.0%	60.0%	8.0%	$1,600.00

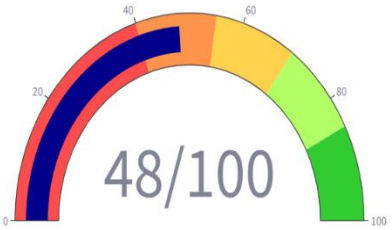
Financial health score

48/100

Health grade

D

Stability status: Stressed / High risk



Budget breakdown

Income vs. Debt payments

AI Financial Advisor

Navigate to:

Risk analysis

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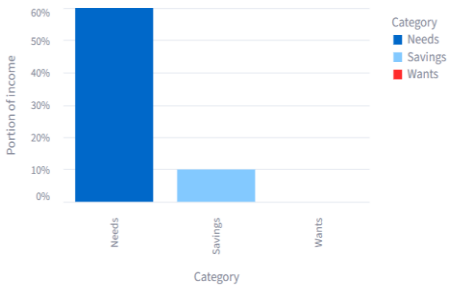

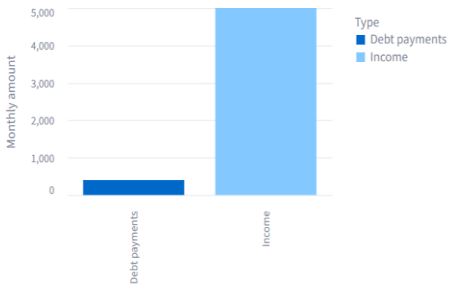

Emergency fund & risk profile

For **6 months** of expenses, your recommended emergency fund is approximately **18,000.00**.

Risk score: **50/100** (Medium). Balanced tolerance for risk and return.

How is my score calculated?

Suggestions to improve your financial health

AI Financial Advisor

Navigate to:

Goal planner

Quick Start Guide

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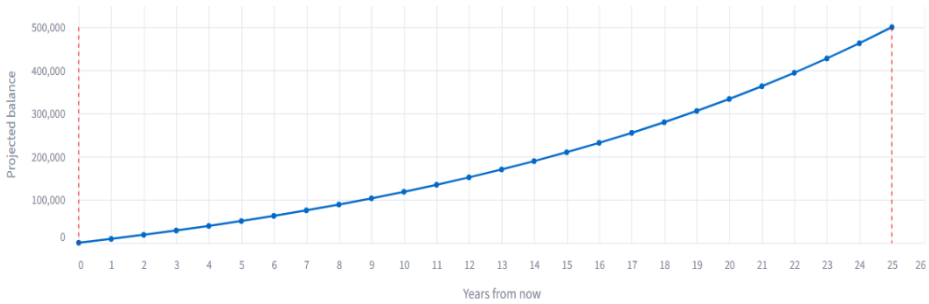
For educational purposes only

Goal Planning & Projections

Plan your financial future with smart projections and scenarios

To reach **Retirement** in **25.0 years**, you should aim to save about **721.51 per month**, assuming an annual return of 6.0%.

Goal projection



AI Financial Advisor

Navigate to:

Goal planner

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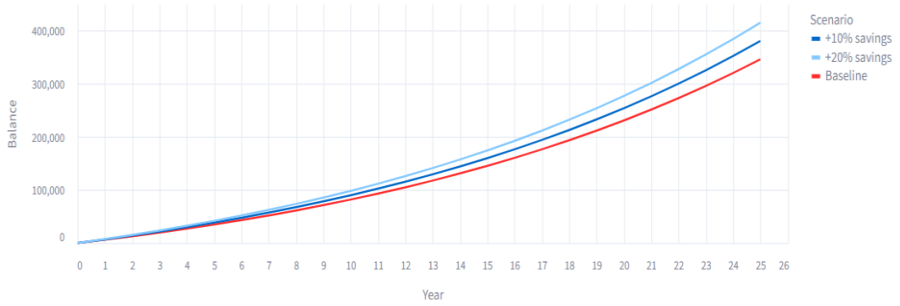
What if I save more?

Predictive savings projections by scenario



AI Financial Advisor

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Ask questions about your finances and get AI-powered advice

You: HOW INCREASE MY SALARY AND SAVE MY MONEY

AI Assistant (Offline Mode): Based on your monthly salary of $5,000.00, here's my recommendation:

Recommended Savings: $1,000.00 (20% of income)

Your current savings rate is 10.0%. Here's a breakdown:

Savings Guidelines: • **Emergency Fund**: 3-6 months of expenses • **Retirement**: 15-20% of income • **Other Goals**: 5-10% of income

Action Steps:

1. Build emergency fund first ($18,000.00)
2. Increase savings gradually to 20%
3. Automate savings transfers

Tip: Start with what you can afford, then increase by 1% every few months.

Ask a question about your finances...

8. ADVANTAGES & DISADVANTAGES

Advantages

• Provides personalized financial advice

• Easy to access from anywhere

• Reduces dependency on human advisors

• Improves financial decision making

• Saves time and cost

Disadvantages

• Advice accuracy depends on input data

• Limited to predefined financial models

• Cannot fully replace professional financial advisors

9. CONCLUSION

The **AI Financial Advisor** project demonstrates how artificial intelligence can assist individuals in managing their finances effectively. The system provides automated financial recommendations based on user data, helping users improve budgeting, savings, and investment planning.

The project successfully shows the potential of AI in financial technology and highlights how digital platforms can make financial advisory services more accessible and efficient.

10. FUTURE SCOPE

Future improvements for the system may include:

• Integration with bank accounts

• Real-time financial tracking

• Advanced AI models for better recommendations

• Mobile application development

• Integration with investment platforms

These enhancements would make the platform more powerful and useful for users.

11. APPENDIX

GitHub & Project Demo Link

Project Website:

https://ai-financial-advisor-mqpu.onrender.com